Compatible tori and a missing Suzuki geometry

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Abstract

A common binary matrix algebra places two nonsplit tori in compatible subfield groups. The resulting rank-three geometry has maximal parabolics Sz(8), Sz(32) and $C_{533} : C_4$, and is absent from the stated exhaustive classification. We rework the previously reviewed explicit example, replacing the finite intersection census by a coprime-order argument. A separate corollary of the scalar-subspace construction contradicts the same source's rank-two uniqueness assertion. Neither argument gives a corrected complete classification.

1 Statement and retained inputs

For subgroups G_i of G, the elements of the coset pregeometry are the labelled left cosets gG_i; incidence is nonempty intersection. Write $G_J = \bigcap_{j\in J} G_j$ and $G_\varnothing = G$. The definition on printed page 46 of [1] calls this system residually weakly primitive if for every proper subset J of the type set some unused i makes $G_{J\cup\{i\}}$ a maximal proper subgroup of G_J. We use exactly this definition.

Theorem 1. *In $G = \mathrm{Sz}(32768)$ there are standard subfield subgroups $S = \mathrm{Sz}(8)$, $T = \mathrm{Sz}(32)$ and a subgroup $K = C_{533} : C_4$ with*

$$S \cap T = C_5 : C_4, \quad S \cap K = C_{13} : C_4, \quad T \cap K = C_{41} : C_4, \quad S \cap T \cap K = C_4. \tag{1}$$

Their rank-three coset geometry is residually weakly primitive, thick, flag transitive and residually connected. Its intersection lattice is absent from the applicable cases of Theorem 5.5 of [1].

The standard faithful Suzuki matrices are denoted by $U(a, b)$, $h(t)$ and w, where w is the antidiagonal identity matrix. Their formulas, Bruhat forms and exact coefficient-field intersections are given in [2, Lemma 1]. In particular, for standard embeddings, $\mathrm{Sz}(2^a) \cap \mathrm{Sz}(2^b) = \mathrm{Sz}(2^{\gcd(a,b)})$, including the degree-one group $\mathrm{Sz}(2) = C_5 : C_4$. We retain the classical Suzuki order formula

$$|\mathrm{Sz}(q)| = q^2(q^2 + 1)(q - 1), \tag{2}$$

maximality of simple subfield groups of prime relative field degree, and maximality of the full normalizers of nonsplit cyclic tori of orders $q \pm \sqrt{2q} + 1$. Such a normalizer has order four times the torus order. These structural facts are among the inputs used by [1, Section 3]; completeness of its geometry lists is not used here. Our common proof [2, Lemma 1], rather than a new subfield-intersection calculation, is an explicit prerequisite.

2 One algebra, two compatible tori

All displayed integer entries in this section encode binary polynomial coefficients in

$$F = \mathbb{F}_2[z]/(z^{15} + z + 1).$$

Thus, for example, 5682 denotes $\sum_{i=0}^{14} \epsilon_i z^i$ with binary digits ϵ_i of 5682; it is not an integer reduced modulo 32768. The polynomial is irreducible. The supplied arithmetic control also checks this directly: successive multiplication by z visits all 32767 nonzero polynomial classes before returning to one.

Put

$$M = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix} = U(1,0).$$

Direct multiplication gives

$$M^4 + M + I = 0, \quad |M| = 15, \quad |r| = 4, \qquad r^{-1}Mr = M^2. \tag{3}$$

The polynomial $X^4 + X + 1$ has no binary root and is not divisible by $X^2 + X + 1$, the only irreducible quadratic over $\mathbb{F}_2$. Consequently $\mathbb{F}_2[M] \cong \mathbb{F}_{16}$. Its scalar extension to F is a field because $\gcd(4, 15) = 1$.

Take $\lambda_8 = 5682$ and $\lambda_{32} = 624$ in the stated encoding, and set

$$a = (M + \lambda_8 I)^{315}, \qquad b = (M + \lambda_{32} I)^{25575}. \tag{4}$$

Here $315 = (8^4 - 1)/13$ and $25575 = (32^4 - 1)/41$. The following table gives their full open Bruhat parameters:

g	u	v	t	x	y	$\lvert g\rvert$
a	893	0	5682	893	893	13
b	5992	4671	1131	5992	1831	41

$$g = U(u,v)h(t)wU(x,y). \tag{5}$$

Equations (3)–(5), the scalar identities $\lambda_8^8 = \lambda_8$, $\lambda_{32}^{32} = \lambda_{32}$, and the indicated orders are finite polynomial identities in F. They are checked by the short witness program supplied with this note, using repeated squaring and the displayed modulus. Each entry of the first parameter row is fixed by eighth powering; each entry of the second is fixed by thirty-second powering. Thus $a \in S$ and $b \in T$. The table is also a direct membership certificate, independently of any enumeration of a Suzuki group.

Since a and b are polynomials in M, they commute. Their distinct prime orders give $|ab| = 13 \cdot 41 = 533$ and $\langle ab \rangle = \langle a, b \rangle$. With $c = ab$, the remaining matrix identity is

$$r^{-1}cr = c^{278}, \qquad 278^2 \equiv -1 \pmod{533}. \tag{6}$$

It follows that $K = \langle c, r \rangle = C_{533} : C_4$: the two cyclic factors have coprime orders, and (6) is an automorphism of order four. Every element has a unique form $a^i b^j r^k$, with $0 \le i < 13$, $0 \le j < 41$ and $0 \le k < 4$. In particular $|K| = 2132$. This K is not the full normalizer of C_{533} in the ambient group; no such maximality is asserted.

Lemma 2. *The intersections are exactly those in (1).*

Proof. The standard-subfield result gives $S \cap T = \mathrm{Sz}(2)$ of order twenty. Both a and r belong to S. If $a^i b^j r^k \in S$, multiplication on the left by a^{-i} and on the right by r^{-k} gives $b^j \in S$. But 41 does not divide $|S| = 29120$. Since b has order 41, we must have $j = 0$. Hence $K \cap S = \langle a, r \rangle$, of order 52. Similarly 13 does not divide $|T| = 32537600$, so $K \cap T = \langle b, r \rangle$, of order 164. Their common intersection contains $\langle r \rangle$ and has order dividing $\gcd(52, 164) = 4$, proving the final equality. The groups of orders 52 and 164 are the full nonsplit torus normalizers in S and T: (6) normalizes their cyclic factors, and the retained normalizer order is respectively $4 \cdot 13$ and $4 \cdot 41$. □

3 The geometry and the source contradiction

We verify all seven instances of residual weak primitivity. The subgroup S is maximal in G because the relative field degree is $15/3 = 5$. Inside S and T, respectively, choose the maximal torus normalizers $S \cap K$ and $T \cap K$. Inside K, choose $K \cap S$, of prime index 41. At each pairwise intersection the final subgroup $\langle r \rangle$ has prime index, respectively 5, 13 and 41. These are all proper type subsets. All eight intersections are distinct, by their orders. This verifies the exact subgroup condition of the source.

For clarity, the geometry properties can be proved without its general rank-three assertion. We use the following elementary observation. If A, B, C are subgroups and

$$A = (A \cap B)(A \cap C), \tag{7}$$

every triple flag of their left-coset system has a common point. Translate the A-coset to A, and write the other two cosets as bB, cC with $b, c \in A$. Factor $b^{-1}c = xy$ with $x \in A \cap B$, $y \in A \cap C$. Then $bx = cy^{-1}$ lies in all three cosets. Every flag of fewer types also extends to a chamber: an incident pair has a common point, and that point supplies the missing coset. Thus (7) gives a geometry and transitivity on every type of flag.

In our example the product set $(K \cap S)(K \cap T)$ has cardinality

$$\frac{52 \cdot 164}{4} = 2132 = |K|.$$

It is contained in K, so equals K. No assumption that an arbitrary subgroup product is a subgroup is needed. Apply (7) with $A = K$. The three panel sizes are 5, 13 and 41, so the geometry is thick.

For residual connectivity, the two residue subgroups in S generate S: one is its maximal torus normalizer of order 52, while the other, $S \cap T$, contains an element of order five and is not contained in that normalizer. The same argument, using order 164, works in T. The product formula gives generation in K. Finally $\langle S, T \rangle = G$, because S is maximal and 31 divides $|T|$ but not $|S|$. The usual coset-graph argument now proves connectedness of the whole incidence graph and of every rank-two residue: a component containing the base flag consists exactly of the cosets reached by words in its incident stabilizer subgroups. Those subgroups generate the relevant group, as just shown. Flag transitivity transports this statement to all residues.

It remains to compare the actual source conclusion. Here the odd field degree is $15 = 3 \cdot 5$, whose prime exponents sum to two. Moreover the subfield $S = \mathrm{Sz}(8)$ has $8 - 1$ prime. The applicable portion of [1, Theorem 5.5, printed pp.58–59] is therefore its first five shapes. Only shape 5 has two Suzuki maximal parabolics and one odd cyclic group extended by C_4. It prescribes the intersection of the two Suzuki parabolics as $C_4 \times C_2$, of order eight; our intersection has order twenty. A permutation of types cannot change this discrepancy, since K is soluble and S, T are not. Shape 8 belongs to the separate branch where the sum of prime exponents exceeds two, and cannot be used in this branch. This proves Theorem 1.

The source's closing qualification on page 61 says that its lists are necessary subgroup lattices, without establishing realizability of every listed shape. Our example contradicts necessity, so that qualification does not remove the counterexample. We do not claim the geometric intersection property. Nor is this rank-three example locally doubly transitive: in the local action on a panel of size thirteen, the stabilizer of one point has order four and cannot be transitive on the other twelve points.

4 A separate rank-two consequence

The preceding example should be distinguished from the locally doubly transitive geometries in [3].

Corollary 3. *The uniqueness assertion for composite $q-1$ in [1, Theorem 6.1, printed p.61] is false at $q = 32768$.*

Proof. Here $q-1 = 32767 = 7 \cdot 31 \cdot 151$ is not prime. The source asserts that the unique residually weakly primitive and locally doubly transitive pregeometry has rank two and parabolics $D_{2(q-1)}$ and $q : (q-1)$, with common subgroup of order $q-1$. Its two valencies are 2 and q.

Apply [3, Theorem 3] to $F = \mathbb{F}_{32768}$ and $K = \mathbb{F}_8$. For each two-dimensional K-subspace T containing K, it gives a rank-two geometry with parabolics $\mathrm{Sz}(8)$ and Q_T, where $|Q_T| = 3584$ and their intersection has order 448. It is residually weakly primitive and locally doubly transitive, with valencies 65 and 8. These differ from 2 and 32768, even after interchanging types. Thus even the underlying incidence geometry is absent from the uniqueness assertion. The forty-one classes counted in [3, Corollary 5] use that paper's stated equivalence under automorphisms of the specified group; a count of forty-one bare incidence isomorphism classes is not needed or asserted here. □

5 Attribution and verification

The binary algebra, fixed scalars, matrices and rank-three construction were obtained in the project's earlier manuscript on compatible Suzuki tori, and were separately reviewed there. We retain that proof, its certificate, original program and review as provenance. The present note reuses the exact witness; its intersection proof uses group orders in place of the earlier enumeration of 2132 elements. The short new replay reads the fixed witness, verifies (3)–(6) and the arithmetic used in Lemma 2, and never searches for M or enumerates an ambient group. Classical Suzuki structure is retained explicitly.

The exact subfield-intersection lemma and the scalar-subspace theorem are acquired from [2, 3]; their complete proofs and reviews accompany this note. The rank-two corollary is a direct use of the latter theorem, not a separate discovery. The earlier higher-rank extension is not a premise. No other source cases or future realization questions are declared settled, and no complete corrected classification, publication priority, new conjecture solution or full Lean proof is claimed.

References

[1] D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63.

[2] Human Compression project, *A missing boundary in the Suzuki subfield construction*, reviewed project manuscript, 2026, Lemma 1.

[3] Human Compression project, *Scalar subspaces in rank-two Suzuki geometries*, reviewed project manuscript, 2026, Theorem 3 and Corollary 5.